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July 2, 2026

Dear Editor,

I am submitting my manuscript **"Causal Transformer: Scaling Gradient-Based Causal Discovery to 500 Variables"** for consideration at *Machine Learning*.

NOTEARS [**?**] is the most widely used differentiable causal discovery framework, but it collapses to zero edges at $d \geq 150$. This has left a gap—$d = 200$ to $d \approx 500$—where no DAG-based method works, even though this is precisely the dimensionality of most real transcriptomic and neuroimaging datasets. My paper fills this gap.

Causal Transformer (CT) treats variables as tokens and learns causal edges from multi-head attention weights, retaining NOTEARS' acyclicity constraint but replacing independent parameter optimization with attention-based W construction. Across 335 experiments—80 synthetic configurations, 15 TCGA real-data seeds at $d = 250$–350, 15 multi-omics seeds, 5 MLP baselines, and extended comparisons with GOLEM and DAGMA—CT is the only gradient-based method that produces non-zero edges at $d \geq 200$. At $d = 200$ on 10 cancer types, CT's top-hub genes are 75% ClinGen Tier-1 cancer drivers. The method runs entirely on a consumer GPU (RTX 5060, 8 GB).

The manuscript includes an Electronic Supplementary Material with full 5-seed sweep tables, ablation results, convergence trajectories, and a DAG-Transformer replication attempt at $d = 200$. A Contribution Information Sheet is attached. A Chinese patent application covering related work has been filed; I declare no other competing interests.

I respectfully suggest Prof. Bryon Aragam (University of Chicago), Prof. Kun Zhang (CMU/MBZUAI), and Prof. Caroline Uhler (MIT/Broad Institute) as potential reviewers.

This work was done independently, without institutional funding or a PhD advisor. I hope the empirical thoroughness speaks for itself.

Sincerely,
Shuaidong Gao